



AI Stock Market Research Assistant

Databricks AI Bootcamp · Capstone Project

Azure Databricks · Meta Llama 3.3 70B · Unity Catalog · Databricks Apps

Medallion Pipeline

Agentic AI (11 tools)

RAG + Vector Search

Databricks Apps

Jayanth Dolai · Senior Software Engineer · Optum / UnitedHealth Group

 github.com/demonjd2026-afk/ai-stock-research-assistant

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Problem Statement

Why build an AI-native stock research platform on the Databricks Lakehouse?



Real-Time Gap

Analysts rely on delayed spreadsheets; no live ingest pipeline for market data into a governed lakehouse.



Fragmented Tooling

Price feeds, news, sentiment, and research notes live in separate silos — no unified platform.



No Agentic Layer

Standard BI dashboards can't answer free-form queries, compare tickers, or take write actions.



News-Price Disconnect

Qualitative news signals are never linked to quantitative price moves in a queryable form.



Solution

An end-to-end Databricks Lakehouse platform ingesting 20 S&P 500 tickers in real time, enriching them with news sentiment via semantic RAG, and exposing a Llama 3.3 70B agentic chat interface for research, comparison, and watchlist management.

Architecture Overview

Medallion Lakehouse · Agentic AI · RAG — fully orchestrated on Azure Databricks



Tickers (20)

AAPL · GOOGL · META · MSFT · NVDA (Tech) | BAC · GS · JPM · MS · V (Finance)
ABBV · JNJ · MRK · PFE · UNH (Healthcare) | AMZN · TSLA · WMT · CVX · XOM

Bronze → Silver

Explicit float()/int() casting · MAX(snapshot_date) dedup · MERGE upsert on (ticker, snapshot_date)
Unity Catalog MODIFY grants (UC metastore v1.0) · News dedup by headline hash

Gold Layer

sector_rankings: SIC descriptions resolved via ticker_config view
top_movers: daily % return ranked per sector · Sentiment aggregates joined to price

AI Agent (11 tools)

Read: get_price_data · get_sentiment · compare_tickers · get_top_movers · search_news ·
get_sector_rankings · flag_price_moves
Write: add_to_watchlist · remove_from_watchlist · save_research_note · save_analysis_report

RAG Pipeline

90 news articles + 20 company profiles → BGE Large embeddings → Databricks AI Search
stock-assistant-vs Vector Search endpoint (ONLINE) · synced daily by Workflow task 4

Workflow Automation

stock-assistant-daily-pipeline | 4-task DAG: bronze → silver → gold → sync_vs_index
Cron: 0 0 22 ? * MON-FRI | Serverless compute — no cluster startup latency

Medallion Pipeline Deep Dive

Bronze → Silver → Gold: data quality, dedup, and aggregation at each layer



Bronze Layer

- Polygon REST API (api.polygon.io)
- 20 tickers across 5 sectors
- Explicit `float()/int()` casting
- Delta Lake partitioned by `snapshot_date`
- Auto-schema detection; ACID writes



Silver Layer

- MERGE upsert on (`ticker`, `snapshot_date`)
- `MAX(snapshot_date)` dedup pre-Gold join
- News dedup by headline hash
- UC grants: MODIFY (metastore v1.0)
- Sentiment linked to ticker × date



Gold Layer

- `sector_rankings`: SIC → clean sector names
- `top_movers`: daily % return ranked
- Sentiment avg joined to price data
- Serves Agent tools + VS sync task
- Optimized for low-latency queries

Agentic AI — 11 Function-Calling Tools

Llama 3.3 70B with a 5-round tool-use loop — reads, writes, and reasons over the lakehouse



get_price_data

Fetch latest price, volume, % return for a ticker from Gold



get_sentiment

Retrieve avg news sentiment score for a ticker



compare_tickers

Side-by-side price + sentiment comparison of 2 tickers



get_top_movers

Top 5 gainers and losers for the day from Gold



search_news

RAG over 110 embedded docs via Databricks AI Search



get_sector_rankings

Sector market cap + avg return from Gold sector_rankings



flag_price_moves

Alert if a ticker moved > N% in a day



add_to_watchlist

Write: add ticker to a named watchlist in Silver



remove_from_watchlist

Write: remove ticker from a watchlist in Silver



save_research_note

Write: persist freeform research note to Delta



save_analysis_report

Write: save structured AI analysis report to Delta

RAG Pipeline — Semantic Search on Financial News

BGE Large embeddings + Databricks AI Search + Llama 3.3 70B grounded responses

Data Sources

90 news articles
20 company profiles
Silver Delta table (news_for_search_index)
110 total documents embedded

Embeddings

BGE Large model
stock-assistant-vs endpoint
Databricks Vector Search (ONLINE)
Auto-sync via Workflow task 4

Query Flow

User query → search_news tool
Semantic similarity search
Top-K relevant chunks retrieved
Context injected into Llama prompt

Grounded Response

Llama 3.3 70B reasons over retrieved context
News grounding prevents hallucination
Source attribution in response
Fallback: direct tool data if RAG scores low

App Demo — Market Summary + Watchlist

Right panel: 20 tickers with % change + named watchlists updated via AI chat · Databricks Apps deployment

AI Stock Market Research Assistant

Powered by Databricks Lakehouse & Meta Llama 3.3 70B

SUGGESTIONS

🍏 Apple stock

🏆 Top movers

⚡ MSFT vs NVDA

📰 AI stocks news

📄 Add to watchlist

🏠 Sector view

📈 Notable moves

💡 Best pick today

Ask me anything about stocks

Press Enter or ↵ to send

Message Stock Assistant...

⬆

🗑

MARKET SUMMARY

Refresh

WATCHLIST

Refresh

Market Summary Sidebar

20 tickers with live price + % change. Green dot = gainer, red dot = loser — sorted by daily return.

Watchlist Panel

Named watchlists (My Watchlist, Tech Watchlist) — populated via agent write tools in chat.

Suggestion Buttons

8 pre-built queries: Apple stock, Top movers, MSFT vs NVDA, AI stocks news, Sector view, and more.

 github.com/demonjd2026-afk/ai-stock-research-assistant

App Interface — Empty State

3-column layout: suggestions panel · chat area · market summary + watchlist sidebar

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💡 Best pick today

Ask me anything about stocks

Press Enter or ↵ to send

Message Stock Assistant...

⬆

🗑

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GOOGL \$377.65 (+2.56%)

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Refresh

📁 WATCHLIST

• AAPL — Tech Watchlist

• AAPL — My Watchlist

• MRK — My WatchList

• MSFT — Tech Watchlist

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• NVDA — My Watchlist

Refresh

Market Summary Loaded

20 S&P 500 tickers displayed with real-time price and % change indicators



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- 📈 Notable moves
- 💡 Best pick today

Show me sector rankings and market cap breakdown

Based on the sector rankings and market cap breakdown, the top 3 sectors by total market cap are:

1. SERVICES-COMPUTER PROGRAMMING, DATA PROCESSING, ETC. with a total market cap of \$6,071.64 billion and an average daily return of 1.8343%.
2. SEMICONDUCTORS & RELATED DEVICES with a total market cap of \$5,005.03 billion and an average daily return of 0.3029%.
3. ELECTRONIC COMPUTERS with a total market cap of \$4,428.17 billion and an average daily return of 2.1984%.

The sector with the highest average daily return is RETAIL-VARIETY STORES with a return of 2.3348%, while the sector with the lowest average daily return is HOSPITAL & MEDICAL SERVICE PLANS with a return of -1.9263%.

The sector with the highest average sentiment score is SEMICONDUCTORS & RELATED DEVICES and SERVICES-BUSINESS SERVICES, NEC with a score of 1.0, indicating a highly positive sentiment.

These rankings and metrics can be useful for investors and analysts to identify trends and opportunities in different sectors.

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MARKET SUMMARY

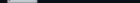
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 Refresh

Best Stock Combination Query

Agent calls compare_tickers + get_sentiment tools — cross-references price gain and sentiment

SUGGESTIONS

🍏 Apple stock

🏆 Top movers

⚡ MSFT vs NVDA

📰 AI stocks news

📁 Add to watchlist

📊 Sector view

📈 Notable moves

💡 Best pick today

Based on the comparison of the stocks, MSFT (Microsoft Corp) has the best combination of price gain and positive sentiment, with a daily return of 2.4766% and an average sentiment score of 0.2. However, it's worth noting that AMZN (Amazon.Com Inc) and NVDA (Nvidia Corp) have a higher average sentiment score of 1.0, indicating a highly positive sentiment, but their daily returns are lower at 0.4926% and 0.3029% respectively.

Today's top gainers are:

1. GOOGL (Alphabet Inc. Class A Common Stock) with a daily return of 2.561%

2. MSFT (Microsoft Corp) with a daily return of 2.4766%

3. WMT (Walmart Inc. Common Stock) with a daily return of 2.3348%

4. AAPL (Apple Inc.) with a daily return of 2.1984%

5. V (VISA Inc.) with a daily return of 2.1644%

Today's top losers are:

1. UNH (UNITEDHEALTH GROUP INCORPORATED) with a daily return of -1.9263%

2. ABBV (ABBVIE INC.) with a daily return of -1.1915%

3. MRK (Merck & Co., Inc.) with a daily return of -0.3581%

What are today's top gainers and losers?

Message Stock Assistant...

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🔄 Refresh



 **Apple stock**

 Top movers

⚡ MSFT vs NVDA

 AI stocks news **Add to watchlist** Sector view

Notable moves

💡 Best pick today

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What are today's top gainers and losers?

Remove MRK from My Watchlist

MRK (Merck & Co., Inc.) has been successfully removed from your watchlist, "My Watchlist".

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- NVDA — My Watchlist

 Refresh

Remove from Watchlist — Write Action

Agent calls remove_from_watchlist tool — watchlist updated and confirmed in chat

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3. MRK (Merck & Co., Inc.) with a daily return of -0.3581%

MRK (Merck & Co., Inc.) has been successfully removed from your watchlist, "My Watchlist".

Remove MRK from My Watchlist

Add UNH to my Watchlist

UNH (UnitedHealth Group Incorporated) has been successfully added to your watchlist, "My Watchlist".

Message Stock Assistant...

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Remove & Add Watchlist — Chained Write Actions

Agent chains `remove_from_watchlist` + `add_to_watchlist` in a single session

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MRK (Merck & Co., Inc.) has been successfully removed from your watchlist, "My Watchlist".

Remove MRK from My Watchlist

Add UNH to my Watchlist

UNH (UnitedHealth Group Incorporated) has been successfully added to your watchlist, "My Watchlist".

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Refresh

Lakebase + Delta CDF

OLTP write-back layer + Change Data Feed — design decisions on Free Edition

Lakebase (Design Intent)

- Lakebase = Databricks-managed Postgres OLTP layer
- Designed for transactional write-back (watchlists, notes)
- OAuth JWT federation — not available on Free Edition
- Architecture pattern documented for production readiness
- Agent write tools target Silver Delta as substitute

Delta CDF (Active)

- Delta Change Data Feed enabled on Silver tables
- Captures INSERT/UPDATE/DELETE with `_change_type` flag
- Streams incremental changes without full table scan
- Powers near-real-time Gold refresh on data change
- Production: CDF → streaming job → Gold

Production Design Intent

- Lakebase: watchlist + notes in Postgres (low-latency OLTP)
- CDF: streaming propagation Silver → Gold deltas
- Free Edition constraint → Delta-only implementation
- Architecture is production-extensible — swap in Lakebase
- Demonstrates awareness of Databricks storage tiers

Workflow Automation

stock-assistant-daily-pipeline — 4-task DAG on Databricks Workflows



Schedule

Cron: 0 0 22 ? * MON-FRI
Runs at 22:00 UTC every weekday
Serverless compute (no cluster startup cost)

Task Dependencies

bronze → silver → gold → sync_vs_index
Linear DAG with dependency declarations
Fail-fast on any upstream error

Compute

Serverless compute for all tasks
No cluster configuration or startup latency
Cost-optimised for daily batch workloads

Monitoring

Task-level status in Workflows UI
Failed run alerts configurable via email
Run history retained for 60 days

Technical Decisions & Learnings

Key engineering trade-offs and constraints solved during implementation

API Selection

api.polygon.io chosen — api.massive.com blocked by Free Edition outbound filter. Polygon provides identical market data with reliable REST endpoints.

Type Casting

Polygon returns inconsistent int/float types. Explicit float()/int() casting in Bronze prevented Silver schema conflicts and downstream Gold aggregation errors.

Databricks SDK vs OpenAI SDK

DATABRICKS_TOKEN is empty at module import time inside Databricks Apps. Switched to get_w().api_client.do() for runtime credential resolution — no hardcoded secrets.

Delta CDF vs Lakebase CDF

Lakebase OAuth JWT federation unavailable on Free Edition. Delta CDF with _change_type flag replicates the CDC pattern with native Delta capabilities.

Gradio HTML vs Chatbot

gr.Chatbot causes JavaScript errors inside Databricks Apps runtime. Replaced with gr.HTML + gr.State for a custom chat bubble renderer — full control, no JS conflicts.

Unity Catalog Grants

UC metastore v1.0 does not support INSERT as a table privilege. MODIFY used instead — documented in setup guide for future team members.

Skills Demonstrated

What this capstone proves as a Senior Software Engineer portfolio piece

Medallion Architecture

Bronze→Silver→Gold pipeline with dedup, MERGE upsert, and partitioned Delta tables — production patterns.

Unity Catalog Governance

Schemas, secret scopes, MODIFY grants, ticker registry — data governance on Free Edition UC.

Semantic RAG

BGE Large embeddings + Databricks AI Search — cutting-edge NLP applied to financial documents.

Agentic AI (11 tools)

Llama 3.3 70B + function-calling + 5-round tool loop — multi-step AI reasoning over the lakehouse.

Workflow Orchestration

Multi-task DAG, cron scheduling, serverless compute — production automation from day one.

Full-stack Deployment

Live Databricks App, Gradio 3-column UI, chat interface, real market data — demo-ready.



github.com/demonjd2026-afk/ai-stock-research-assistant



linkedin.com/in/jayanth-dolai-7b115213a

Jayanth Dolai — Senior Software Engineer